

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B Tech-AI&DS	Date:	12-08-2026

Code of Conduct

I Y Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y Greeshma Reddy

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:

Incorrect interpretation of ambiguous sentences

Failure to handle subject–verb agreement

Inefficient parsing for long queries

Example user query:

“Show me the transactions with the card from last month”

a) Analyze the ambiguity present in the given sentence using CFG

b) Evaluate the limitations of CFG in handling real-world conversational queries

c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing

d) Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:

Backtracking issues

Poor handling of incomplete or ambiguous input

Difficulty in real-time response

Example command:

“Book a flight to Delhi with a window seat”

a) Analyze how the given sentence can lead to multiple parse structures

b) Analyze the limitations of top-down parsing in this real-time application

c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs

d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.

The system must handle:

Complex and lengthy medical sentences

Ambiguity in sentence structure

Accurate subject-verb agreement

Relationships between medical actions and entities (sub-categorization)

Real-time processing for hospital use

Example input:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.

a) Design a complete NLP architecture integrating:

CFG for syntactic structure

Probabilistic CFG (PCFG) for ambiguity resolution

Feature Structures for agreement handling

Efficient parsing techniques

b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).

c) Create a method to:

Incorporate sub-categorization frames for medical verbs

Handle real-time data processing

Improve accuracy and scalability in a hospital environment

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question 1 – Banking Customer Support Chatbot

1. Understanding of Industry Problem

- Banking queries can contain **ambiguity and complex sentence structures**.
- Example: “Show me the transactions with the card from last month.”
- “with the card” and “from last month” can have different grammatical attachments.
- Basic CFG may not identify the intended meaning correctly.

2. Application of Engineering Concepts

- **CFG**: Provides basic syntactic structure.
- **PCFG**: Assigns probabilities to different parse structures.
- **Feature Structures**: Handle subject–verb agreement.
- **Earley Parsing**: Efficiently handles ambiguous and complex sentences.

3. Solution Design

- Proposed architecture:
 - User Query
 - ↓
 - Tokenization
 - ↓
 - CFG Parsing
 - ↓
 - Feature Checking
 - ↓
 - Earley Parsing
 - ↓
 - PCFG Ranking
 - ↓
 - Semantic Interpretation
 - ↓
 - Banking Response
- Final meaning: **Transactions + Card + Last Month**.

4. Use of Tools

- **Python** for implementation.
- **NLTK** for CFG, PCFG and parsing.
- **spaCy** for NLP preprocessing and entity extraction.
- Banking APIs for retrieving transaction information.

5. Analysis

- CFG alone has difficulty resolving ambiguity.
- PCFG selects the most probable interpretation.
- Feature Structures improve grammatical accuracy.
- Earley Parsing reduces repeated parsing and improves efficiency.

6. Professionalism

- The solution is suitable for real banking chatbots.
- Improves **accuracy, response time and reliability**.
- Can be integrated with real-time banking services.

Question 2 – Voice Assistant

1. Understanding of Industry Problem

- Voice assistants need to process commands quickly.
- Example: **“Book a flight to Delhi with a window seat.”**
- The phrase “with a window seat” may have different syntactic attachments.
- Top-down parsing may face backtracking and delays.

2. Application of Engineering Concepts

- **Top-Down Parsing:** Starts from the start symbol and predicts grammar rules.
- **Earley Parsing:** Uses prediction, scanning and completion.
- Earley parsing can handle **ambiguity and incomplete input** better.
- It avoids unnecessary repeated parsing.

3. Solution Design

- Proposed architecture:
 - Speech Input
 - ↓
 - Speech-to-Text
 - ↓
 - Earley Parser
 - ↓
 - Ambiguity Resolution
 - ↓
 - Intent & Entity Extraction
 - ↓
 - Flight Booking
 - ↓
 - Voice Response

- Output: **Book Flight** → **Delhi** → **Window Seat**.

4. Use of Tools

- **Python** for implementation.
- **NLTK** for parsing.
- Speech-to-text APIs for voice conversion.
- **spaCy** for entity extraction.
- FastAPI for real-time service integration.

5. Analysis

- Top-down parsing can cause excessive backtracking.
- It is less effective for incomplete voice commands.
- Earley parsing handles ambiguous and partial sentences.
- Therefore, Earley parsing is more suitable for real-time voice assistants.

6. Professionalism

- Provides faster and more flexible voice command processing.
- Improves user experience.
- Can be integrated with real-world booking and assistant applications.

Question 3 – Healthcare Medical Report System

1. Understanding of Industry Problem

- Medical reports contain **long, complex and ambiguous sentences**.
- Subject–verb agreement must be handled correctly.
- The system must identify relationships between doctors, patients, treatments and follow-ups.
- Real-time processing is important in hospitals.

2. Application of Engineering Concepts

- **CFG**: Identifies sentence structure.
- **PCFG**: Resolves ambiguous interpretations.
- **Feature Structures**: Handle agreement.
- **Earley Parsing**: Provides efficient parsing.
- **Sub-categorization**: Identifies relationships between medical verbs and entities.

3. Solution Design

- Proposed architecture:
 - Medical Report
 - ↓
 - Preprocessing
 - ↓

- CFG Parsing
- ↓
- Feature Structures
- ↓
- Earley Parsing
- ↓
- PCFG
- ↓
- Medical NER & Sub-categorization
- ↓
- Semantic Extraction
- ↓
- EHR System
- Output:
 - **Treatment:** Start medication
 - **Follow-up:** Schedule visit
 - **Location:** Chennai
 - **Diagnosis:** Not explicitly mentioned

4. Use of Tools

- **Python** for implementation.
- **NLTK** for CFG, PCFG and parsing.
- **spaCy** for medical entity extraction.
- **Pandas** for data processing.
- **FastAPI/Docker** for deployment.

5. Analysis

- CFG alone cannot effectively handle medical ambiguity and relationships.
- PCFG improves ambiguity resolution.
- Feature Structures improve agreement checking.
- Earley Parsing improves processing of complex sentences.
- Sub-categorization improves identification of medical actions and entities.

6. Professionalism

- The proposed architecture is suitable for hospital environments.
- Supports **accuracy, scalability and real-time processing**.
- Can be integrated with Electronic Health Record (**EHR**) systems.
- Reduces errors in extracting important medical information.